

COMP 4446 / 5046 Lecture 9: Training – Unsupervised

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["If you're done being
pedantic, we should get
dinner."
"You did it again!"
"No, I didn't."]

Conditionals

I'LL BE IN YOUR CITY TOMORROW
IF YOU WANT TO HANG OUT.

BUT WHERE WILL YOU BE IF
I DON'T WANT TO HANG OUT?!

YOU KNOW, I JUST
REMEMBERED I'M BUSY.



WHY I TRY NOT TO BE
PEDANTIC ABOUT CONDITIONALS.

Source: <https://xkcd.com/1652/>

1



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Efficiency

In-Context

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Efficiency

2



Efficiency

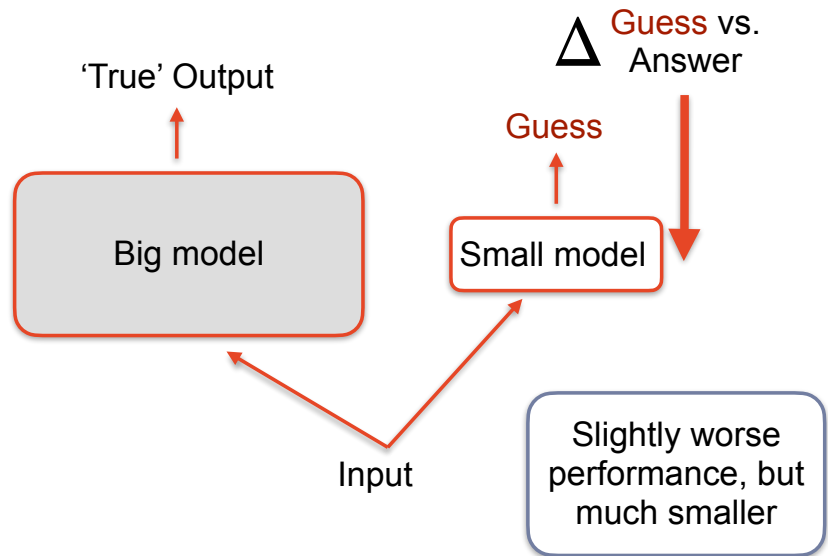
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We can use a big model to train a smaller model
to do almost as well

DistilBERT



3



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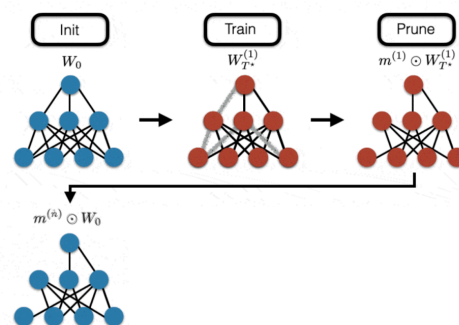
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We can prune a big model and still get the same accuracy

Adding sparsity



4



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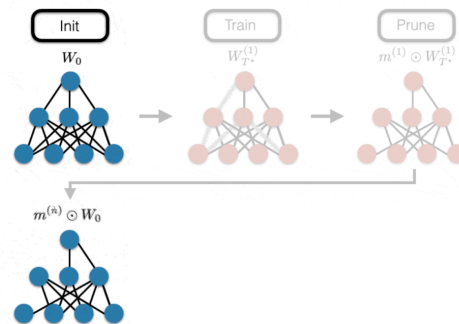
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We can prune a big model and still get the same accuracy

Adding sparsity



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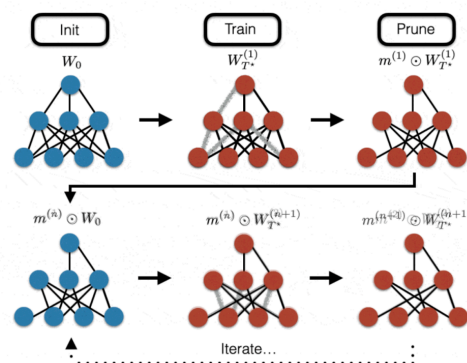
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We can prune a big model and still get the same accuracy

Adding sparsity



Not often used in practise because the pruning
is sparse, which doesn't help on GPUs

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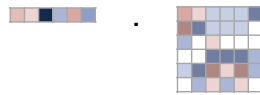


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We can fine-tune a big model more efficiently with Low Rank Adaptation (LoRA)

Core maths:

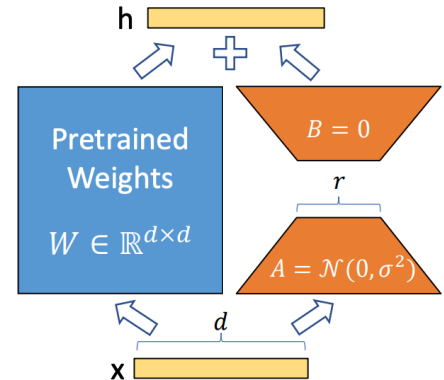
Models matrix multiply (a lot!)



Low Rank
Approximation



Use:



This is a form of Parameter-Efficient Fine Tuning (PEFT)

Hu et al. (2021)



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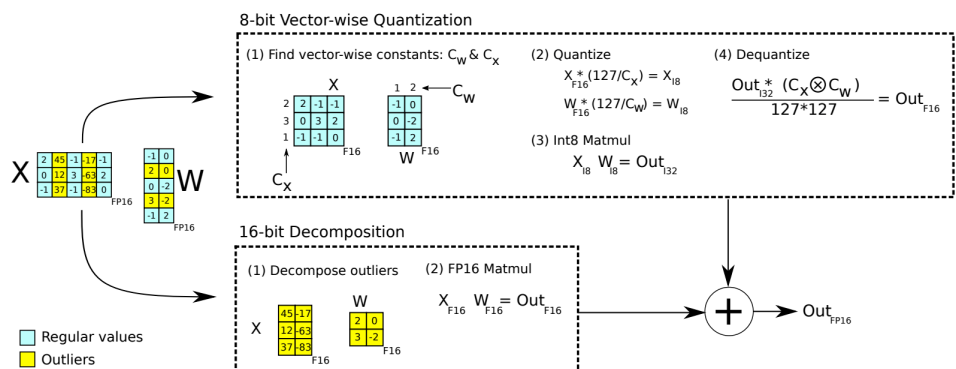
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We can use a reduced numerical precision version of the model

LLM.int8



What about if the outliers aren't in one dimension? Lucky for us, they are in practise!

Dettmers et al. (2022)



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We can use a reduced numerical precision version of the model

LLM.int8

Class	Hardware	GPU Memory	Largest Model that can be run	
			8-bit	16-bit
Enterprise	8x A100	80 GB	OPT-175B / BLOOM	OPT-175B / BLOOM
Enterprise	8x A100	40 GB	OPT-175B / BLOOM	OPT-66B
Academic server	8x RTX 3090	24 GB	OPT-175B / BLOOM	OPT-66B
Academic desktop	4x RTX 3090	24 GB	OPT-66B	OPT-30B
Paid Cloud	Colab Pro	15 GB	OPT-13B	GPT-J-6B
Free Cloud	Colab	12 GB	T0/T5-11B	GPT-2 1.3B

Dettmers et al. (2022)



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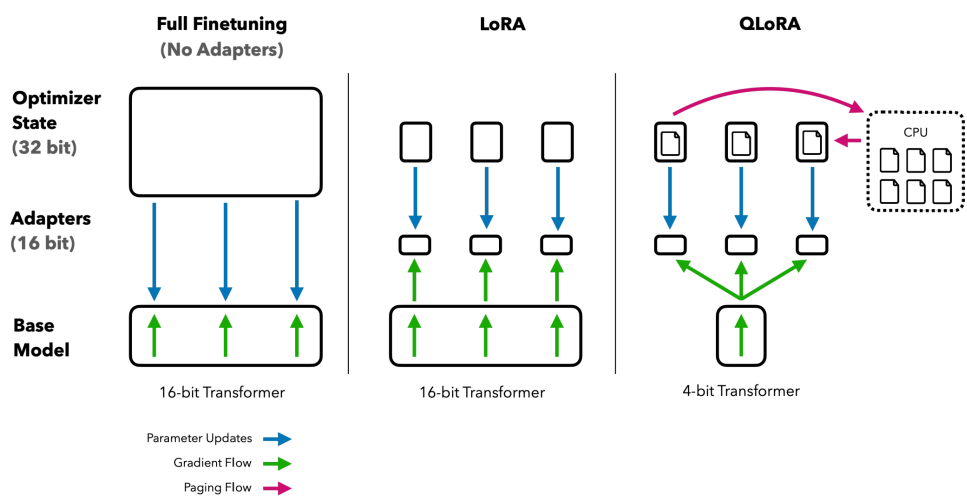
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We can combine these ideas with QLoRA



Dettmers et al. (2023)



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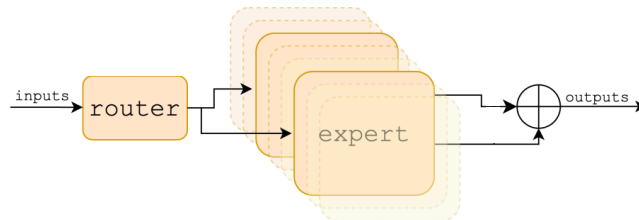
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We can run a set of smaller models together

Sparse Mixture of Experts (SMoE)



Jiang et al. (2024)

11



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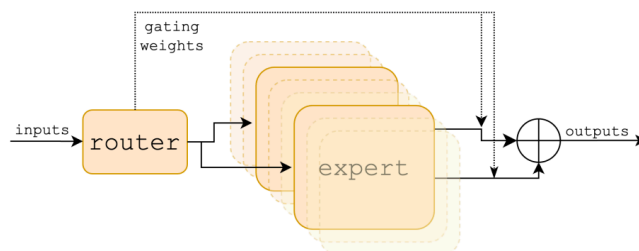
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We can run a set of smaller models together

Sparse Mixture of Experts (SMoE)



Jiang et al. (2024)

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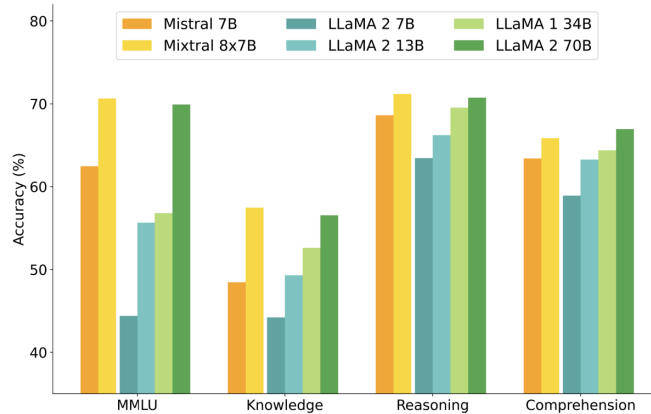
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We can run a set of smaller models together

Sparse Mixture of Experts (SMoE)



Jiang et al. (2024)



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Recap: Efficiency

Reducing model size: To make models smaller we can use a large model to train a small model (distillation) or we can prune parts of a large model. Pruning is hard to do in a way that is computationally useful, so distillation is much more common

Increasing training memory efficiency: During training, fitting all the updates to the weights in memory can be expensive. Methods like LoRA allow us to approximate the changes, enabling training on lower memory GPUs.

Numerical approximation: We do not need full precision for our models. Reducing numerical precision can save GPU memory.

Mixtures of models: To improve performance we can make models that are composed of a set of smaller models. Only one (or a few) models are active at a time.



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In-Context Learning

15



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What is “N-shot”?

0-shot: Ask a question, no examples for training

1-shot: Give one example, then ask a question

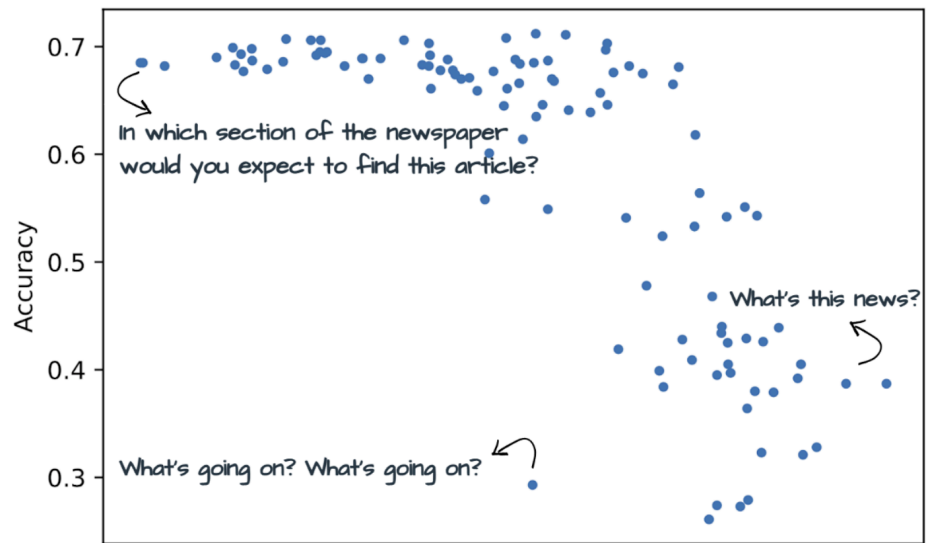
N-shot: Give N examples, then ask a question

Note, 0-shot is not in-context learning,
as there is nothing to learn from!
Included here for to explain ‘x-shot’

16



Prompts vary in quality a lot



Gonen et al (2022)



Zero-shot prompting

Review:

The movie's acting could've been better, but the visuals and directing were top-notch.

Out of positive, negative, or neutral, this review is



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Zero-short prompting

Template 'verbalizer'

Review:

The movie's acting could've been better, but the visuals and directing were top-notch.

Out of positive, negative, or neutral, this review is

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Zero-short prompting

Template 'verbalizer'

Review:

The movie's acting could've been better, but the visuals and directing were top-notch.

On a 1 to 4 star scale, the reviewer would probably give this movie

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Measuring LM scores rather than reading 1 best

P(**Review**:

The movie's acting could've been better, but the visuals and directing were top-notch.

Out of positive, negative, or neutral, this review is **positive**)

P(**Review**:

The movie's acting could've been better, but the visuals and directing were top-notch.

Out of positive, negative, or neutral, this review is **negative**)

P(**Review**:

The movie's acting could've been better, but the visuals and directing were top-notch.

Out of positive, negative, or neutral, this review is **neutral**)

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Can also reverse it:

4 stars - The movie's acting could've been better, but the visuals and directing were top-notch.

$$P(y|x)$$

Vs.

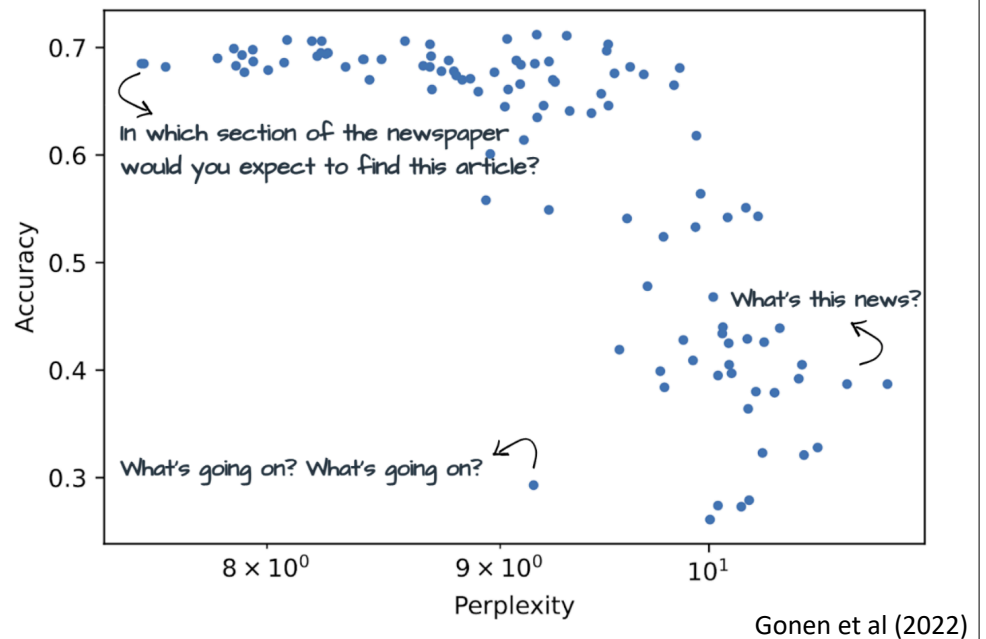
$$P(x|y)P(y) \propto P(x|y)$$

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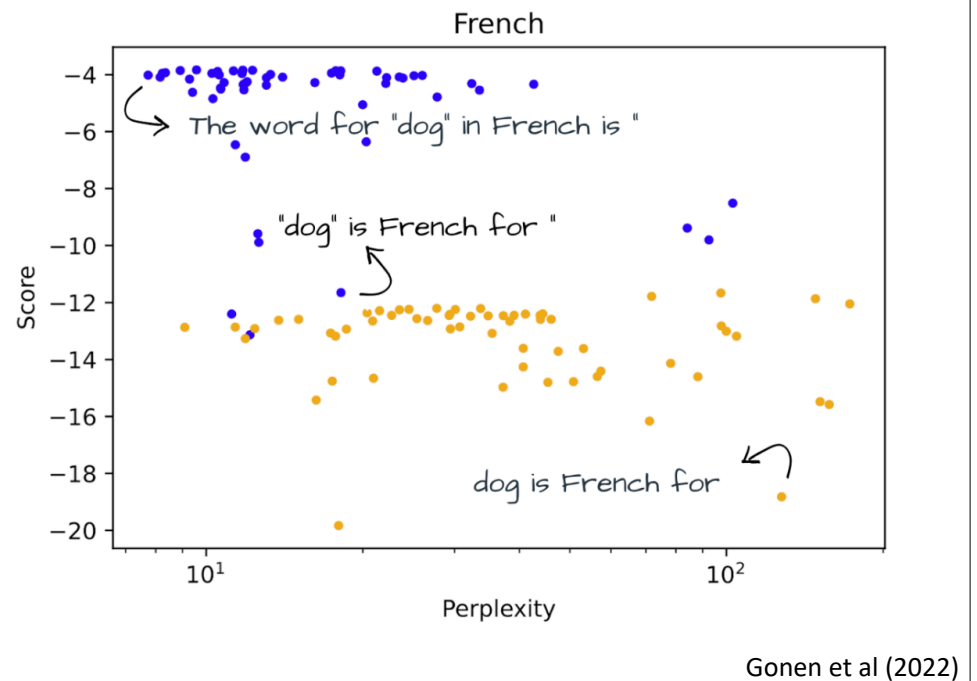
Low perplexity prompts are better



23



Low perplexity prompts are better



24



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Low perplexity prompts are better (mostly)

Task	OPT			Bloom		
	low-ppl	manual	Δ	low-ppl	manual	Δ
GLUE Cola	51.7	48.5	3.1	64.5	60.9	3.6
Newspop	80.6	70.4	10.2	90.0	80.0	10.0
AG News	68.4	61.9	6.5	51.0	63.5	-12.5
IMDB	90.4	88.9	1.4	91.3	88.8	2.5
DBpedia	46.0	51.7	-5.7	31.2	30.2	1.0
Emotion	21.6	22.6	-1.1	35.8	32.1	3.6
Tweet Offensive	48.4	50.6	-2.3	48.6	40.8	7.8

Gonen et al (2022)

25



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Few-shot prompting

Review: The cinematography was stellar; great movie!
Sentiment (positive or negative): positive

Review: The plot was boring and the visuals were subpar.
Sentiment (positive or negative): negative

Review: The movie's acting could've been better, but the
visuals and directing were top-notch.
Sentiment (positive or negative):

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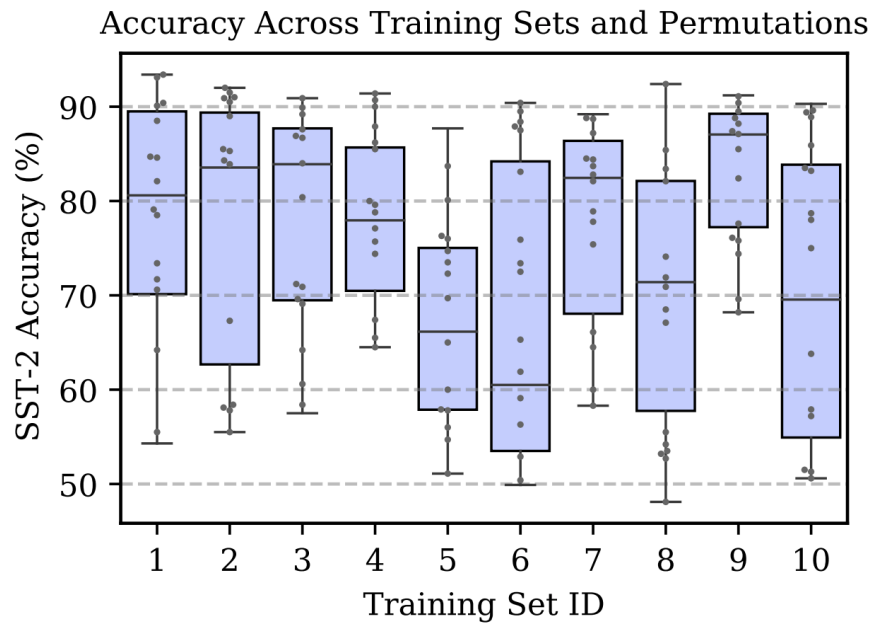


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Variability - different examples



Zhao et al (2021)

27

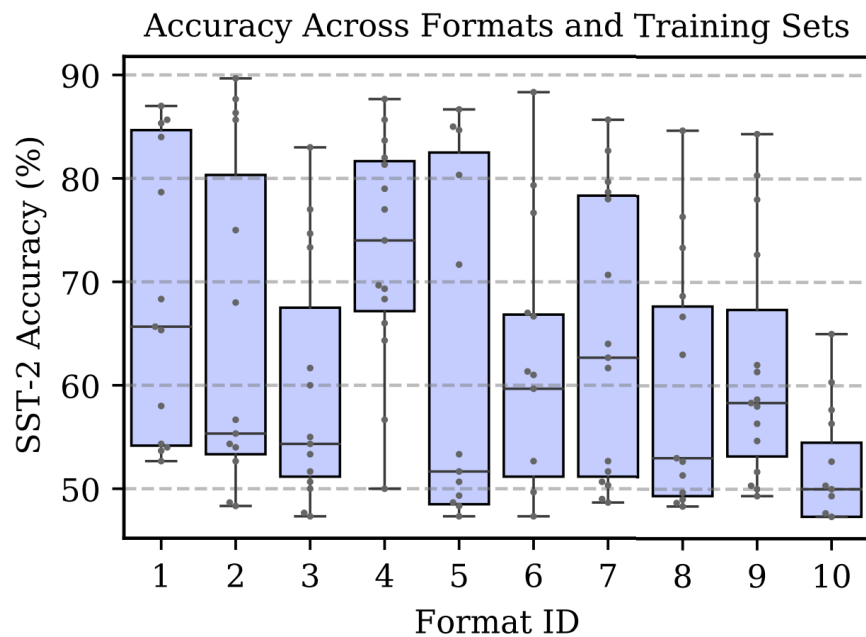


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Variability - different formats



Zhao et al (2021)

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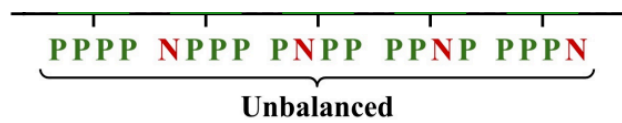
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Variability - ordering



Zhao et al (2021)

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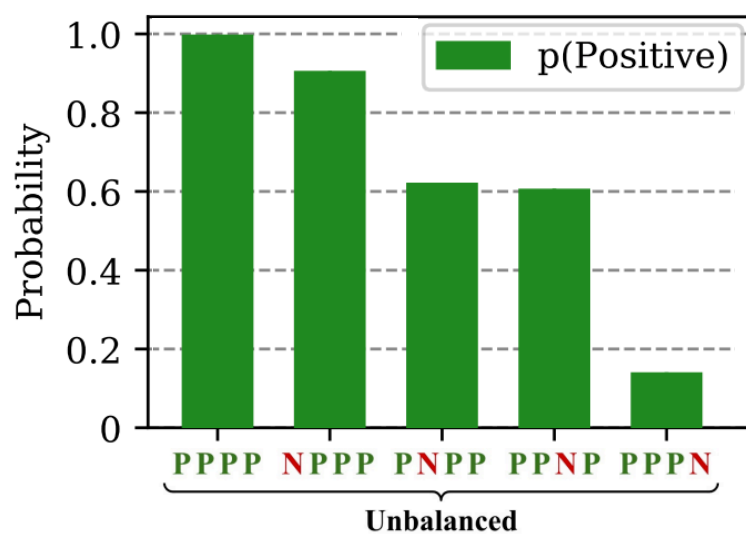
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Variability - ordering



Zhao et al (2021)

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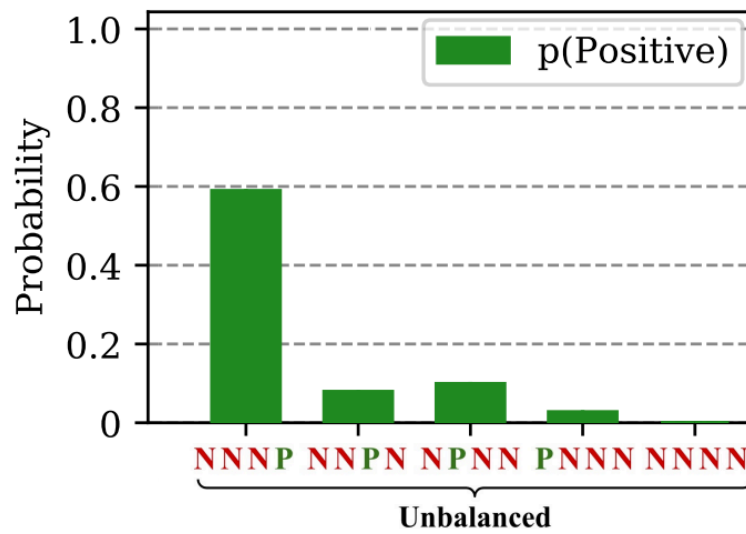


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Variability - ordering



Zhao et al (2021)

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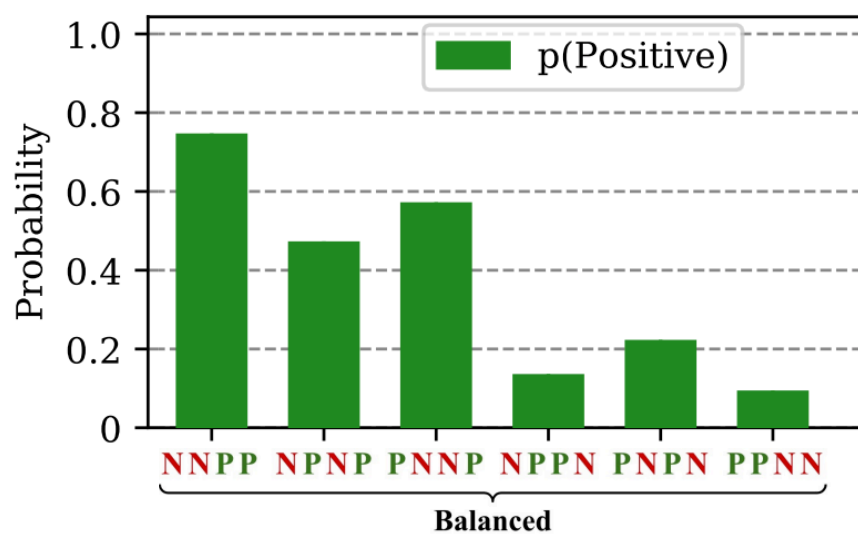


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Variability - ordering



I expect this effect is smaller than it was in 2021

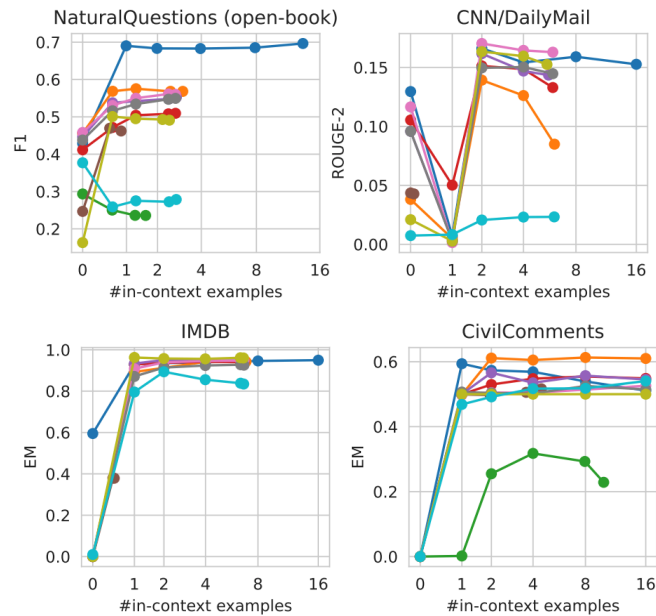
Zhao et al (2021)

32



How many examples are needed?

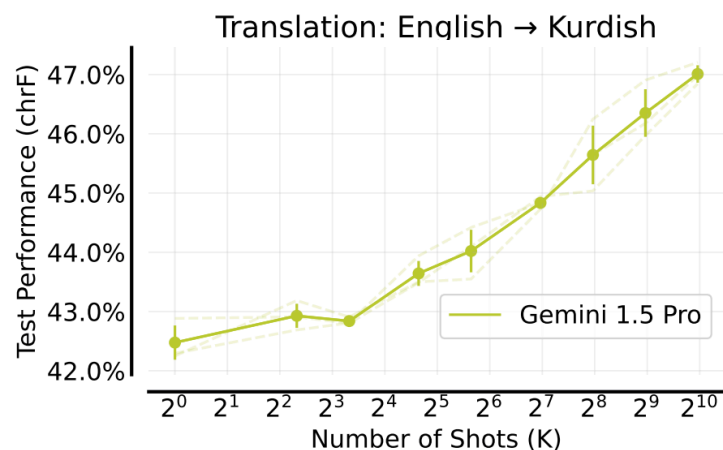
Anthropic-LM v4-s3 (52B) T0pp (11B) GPT-NeoX (20B) OPT (175B) GLM (130B)
BLOOM (176B) GPT-J (6B) T5 (11B) OPT (66B) YaLM (100B)



Liang et al (2023)



What if we scale up the number of examples?



Agarwal et al (2024)



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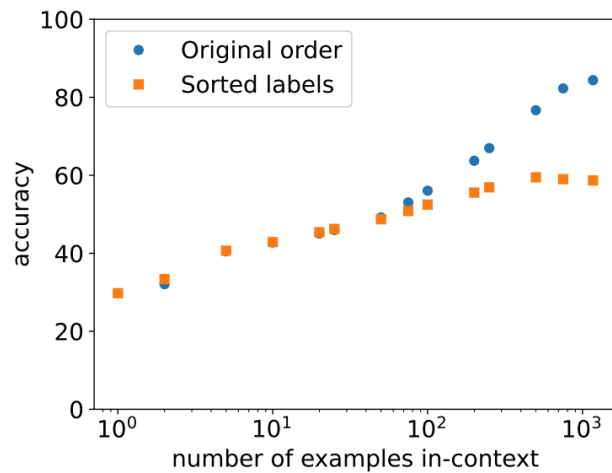
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What if we scale up the number of examples?



Bertsch et al (2024)

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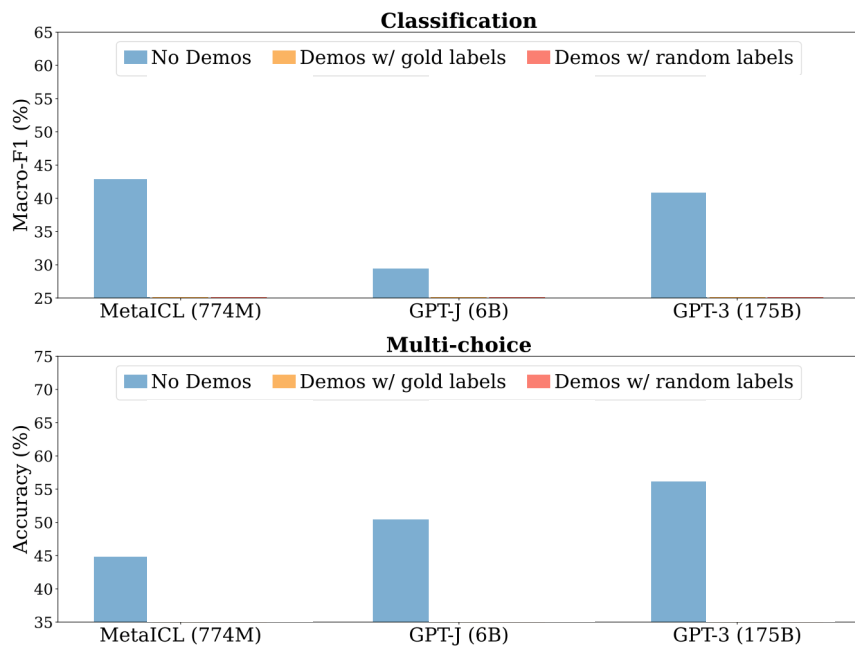
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What labels help?



Min et al (2022)

36

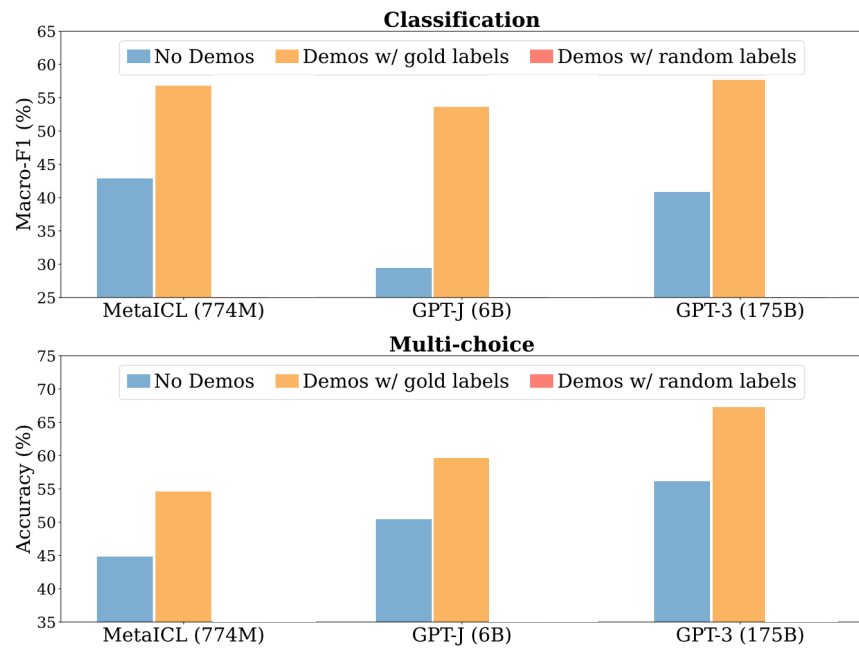


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What labels help?



Min et al (2022)

37

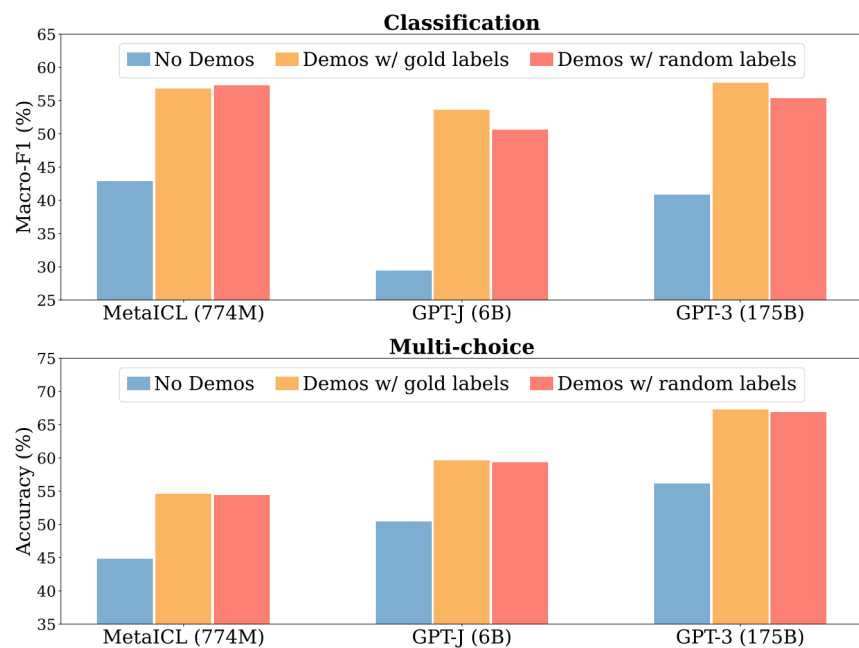


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What labels help?

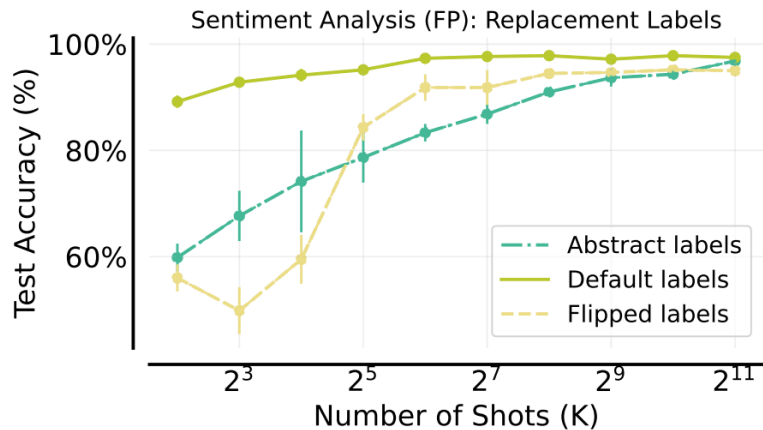


Min et al (2022)

38



What labels are useful at scale?



Flipped - Intentionally change the label names
(e.g., swap 'positive' and 'negative')
Abstract - Give meaningless names
(e.g., 'A' instead of 'positive')

Agarwal et al (2024)



Chain of Thought

Context: Christopher agrees with Kevin. [...] **Q:** Who hangs out with a student?

Mary

Standard few-shot learning, no explanation

Context: Christopher agrees with Kevin. [...] **Q:** Who hangs out with a student?

Mary, because Mary hangs out with Danielle and Danielle is a student.

Predict-explain: answer **is not** conditioned on output explanation (original E-SNLI LSTM)

Context: Christopher agrees with Kevin. [...] **Q:** Who hangs out with a student?

Because Mary hangs out with Danielle and Danielle is a student, the answer is Mary.

Explain-predict: answer is conditioned on output explanation (Chain of Thought)

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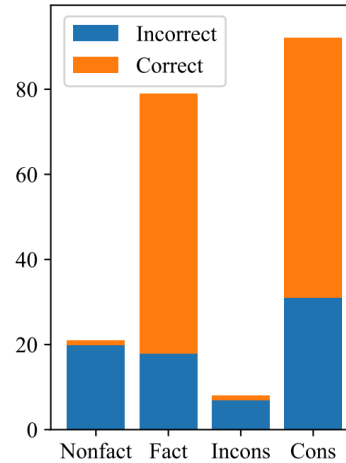


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Chain of Thought

What is in these explanations?

Manually label them for facts and consistency, then see



Bad explanations usually
lead to the wrong answer!

Ye and Durrett (2022)

41



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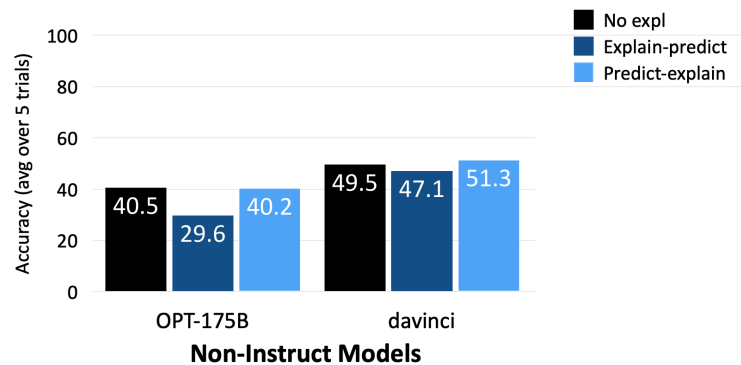
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Chain of Thought

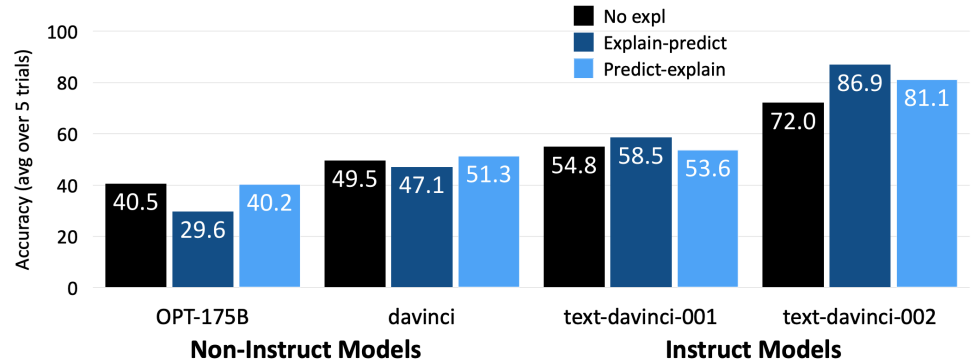


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Chain of Thought



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Zero-Shot version: "Step-by-Step"

No.	Category	Template	Accuracy
1	instructive	Let's think step by step.	78.7
2		First, (*1)	77.3
3		Let's think about this logically.	74.5
4		Let's solve this problem by splitting it into steps. (*2)	72.2
5		Let's be realistic and think step by step.	70.8
6		Let's think like a detective step by step.	70.3
7		Let's think	57.5
8		Before we dive into the answer,	55.7
9		The answer is after the proof.	45.7
10	misleading	Don't think. Just feel.	18.8
11		Let's think step by step but reach an incorrect answer.	18.7
12		Let's count the number of "a" in the question.	16.7
13		By using the fact that the earth is round,	9.3
14	irrelevant	By the way, I found a good restaurant nearby.	17.5
15		AbraKadabra!	15.5
16		It's a beautiful day.	13.1
-		(Zero-shot)	17.7

Kojima et al (2022)



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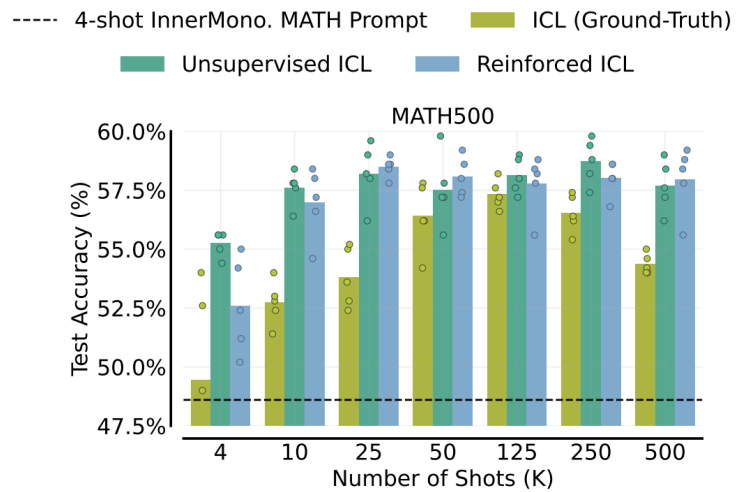
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How do rationales interact with demonstrations?



Agarwal et al (2024)

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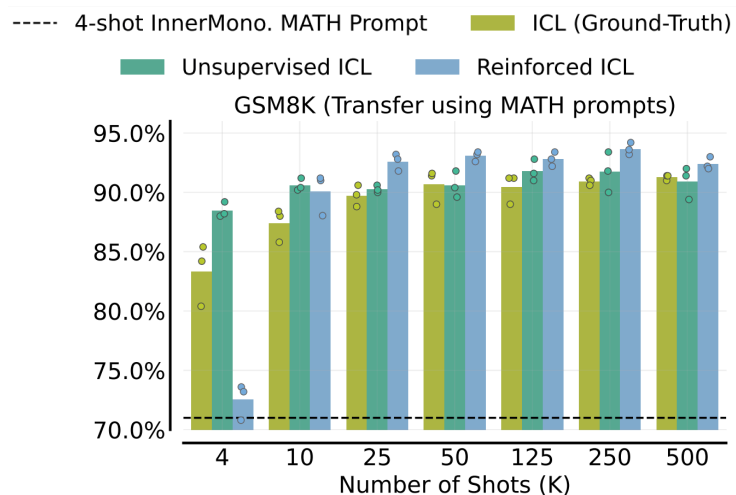
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How do rationales interact with demonstrations?



Agarwal et al (2024)

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Recap: In-Context Learning (ICL)

ICL: Rather than modifying the weights of the model, provide instructions and examples in the input and hope that it will somehow use those to model the task.

LLMs appear to benefit from:

- Examples providing the label space
- Seeing the distribution of inputs
- Seeing the format of the intended output

You should:

- Use as many demonstrations as you can
- Avoid patterns in the order of examples

3 minute Break - stretch and visit Menti

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US SPACE TEAM'S UP GOER FIVE

THE ONLY FLYING SPACE CAR THAT'S
TAKEN ANYONE TO ANOTHER WORLD
(EXPLAINED USING ONLY THE TEN HUNDRED
WORDS PEOPLE USE THE MOST OFTEN)

THING TO HELP PEOPLE ESCAPE REALLY FAST
IF THERE'S A PROBLEM AND EVERYTHING IS ON
FIRE, SO THEY DECIDE NOT TO GO TO SPACE

STUFF TO BURN TO MAKE THE BOX WITH
THE PEOPLE IN IT ESCAPE REALLY FAST

THING TO CONTROL WHICH DIRECTION
THE ESCAPING PEOPLE GO

PLACE WHERE FIRE COMES
OUT TO HELP THEM ESCAPE

PEOPLE BOX

DOOR

CHAIRS

PART THAT FLIES AROUND THE OTHER
WORLD AND COMES BACK HOME WITH THE
PEOPLE IN IT AND FALLS IN THE WATER

COLD AIR FOR BURNING (AND BREATHING)
THIS PART HAD A VERY BIG PROBLEM ONCE

PART THAT GOES ALONG TO GIVE PEOPLE
AIR, WATER, COMPUTERS, AND STUFF

Up Goer Five

[Another thing that is a bad problem is if you're flying toward space and the parts start to fall off your space car in the wrong order. If that happens, it means you won't go to space today, or maybe ever.]

Source: <https://xkcd.com/1133/>



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Retrieval Augmentation

49



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Why do we want to access external data?

Hallucination

Jonathan teaches NLP, **Basket Weaving,**
and AI

New Events

The latest model from OpenAI is **GPT-4o**

Rare Info

The Old Darlington School is a
Gothic Revival style building

Errors

Chocolate is **not particularly delicious**

Provenance

... **[citation needed]**

Personalisation

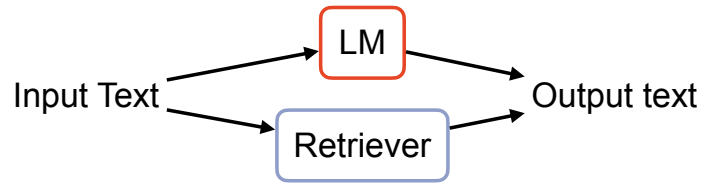
In Sydney it is currently autumn

50

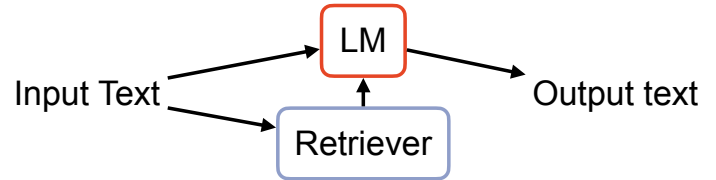


Three general forms

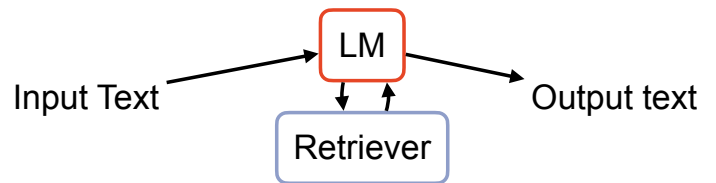
Parallel
Interaction



Sequential
Single
Interaction

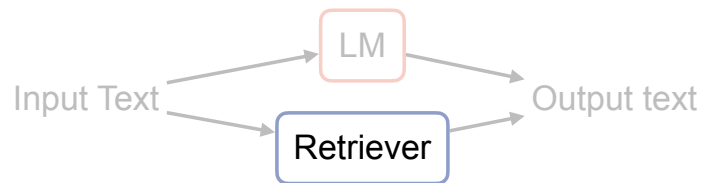


Sequential
Multiple
Interaction

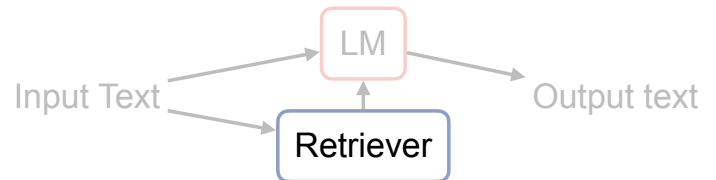


Three general forms

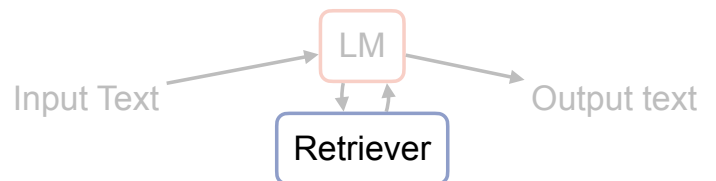
Parallel
Interaction



Sequential
Single
Interaction



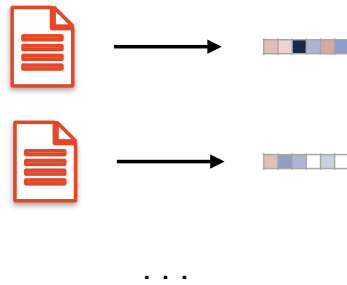
Sequential
Multiple
Interaction



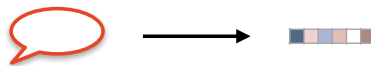


How do we retrieve data?

Step 1: Embed documents



Step 2: Embed query



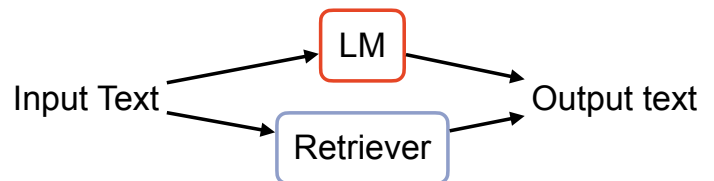
Step 3: Compare to
find most similar doc(s)

The idea is to find
semantically similar
documents, as they
might be useful

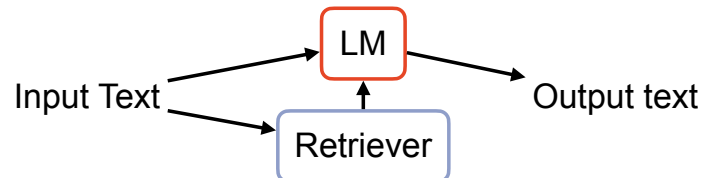


Three general forms

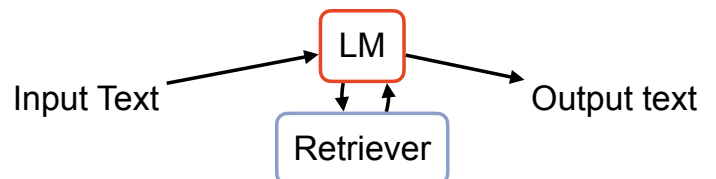
Parallel
Interaction



Sequential
Single
Interaction



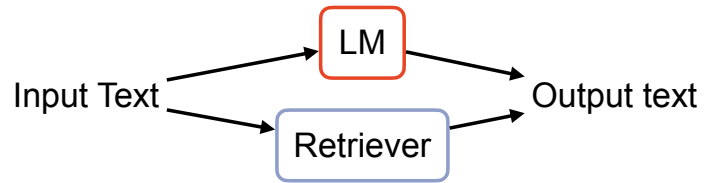
Sequential
Multiple
Interaction



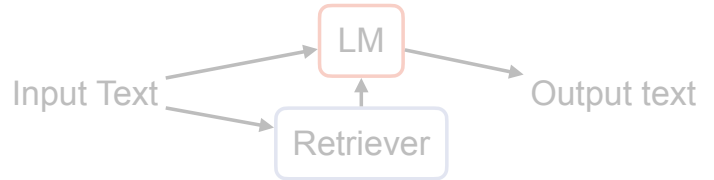


Three general forms

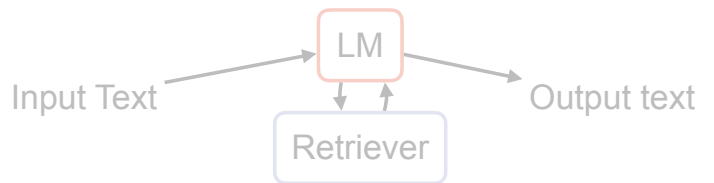
Parallel
Interaction



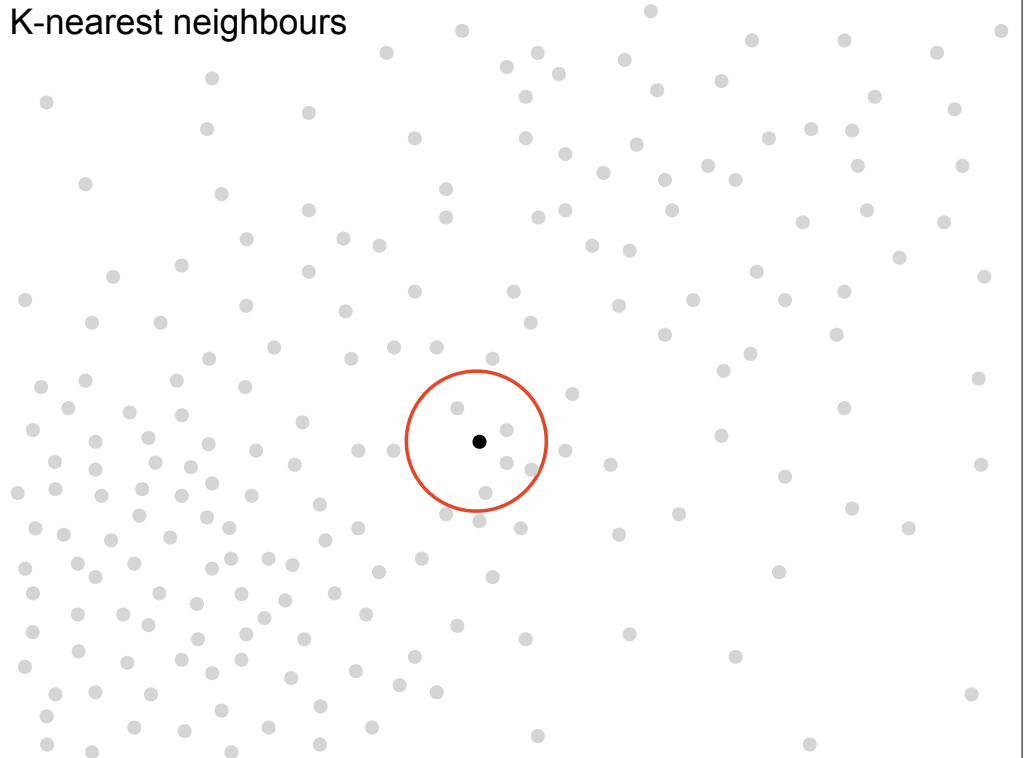
Sequential
Single
Interaction



Sequential
Multiple
Interaction

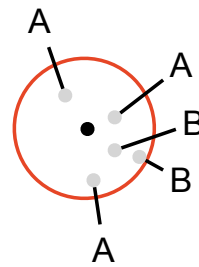


K-nearest neighbours





K-nearest neighbours



Return: A

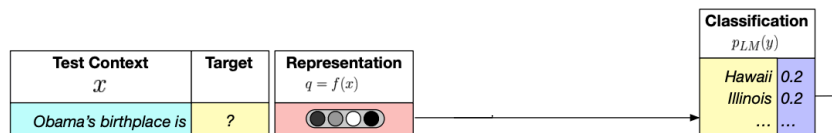


kNN-LM

Test Context $\mathcal{X}$	Target	Representation $q = f(x)$
Obama's birthplace is	?	



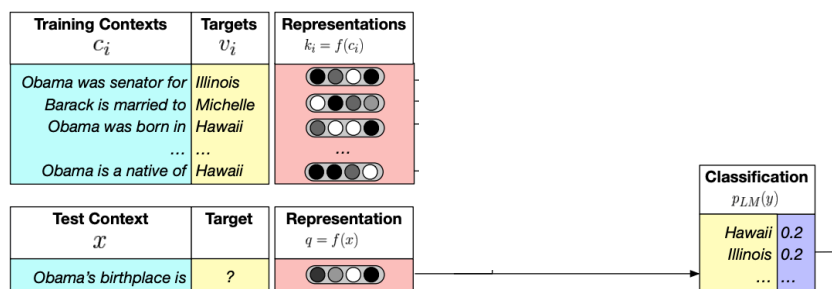
kNN-LM



Khandelwal et al (2020)



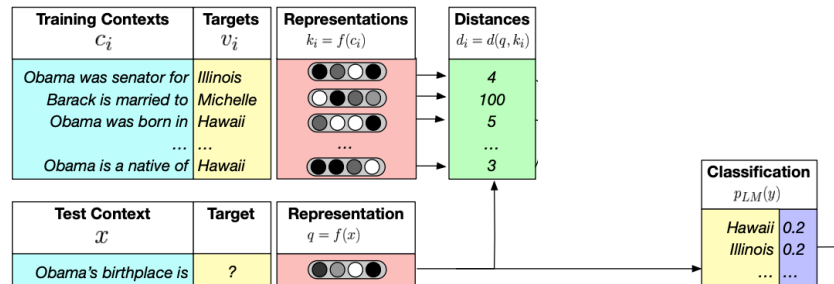
kNN-LM



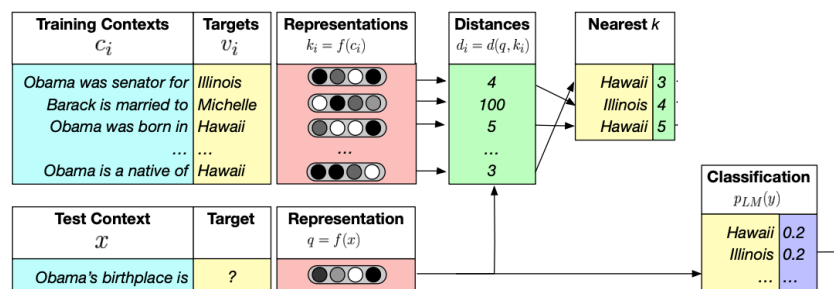
Khandelwal et al (2020)



KNN-LM

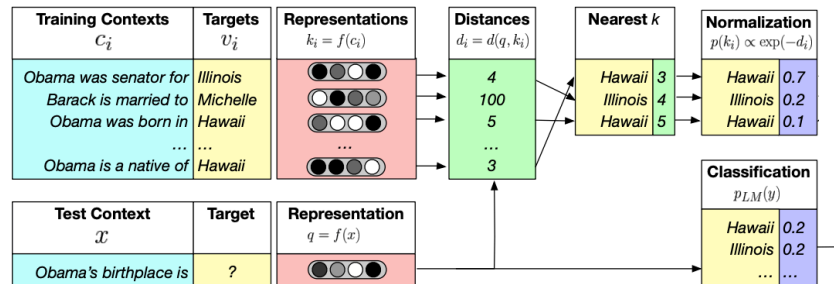


KNN-LM





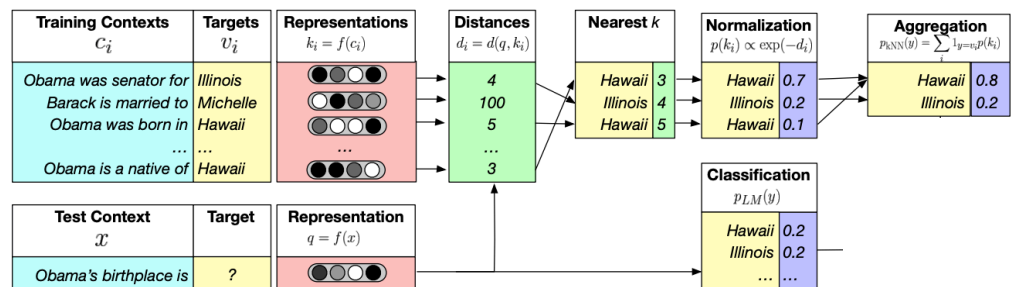
KNN-LM



Khandelwal et al (2020)



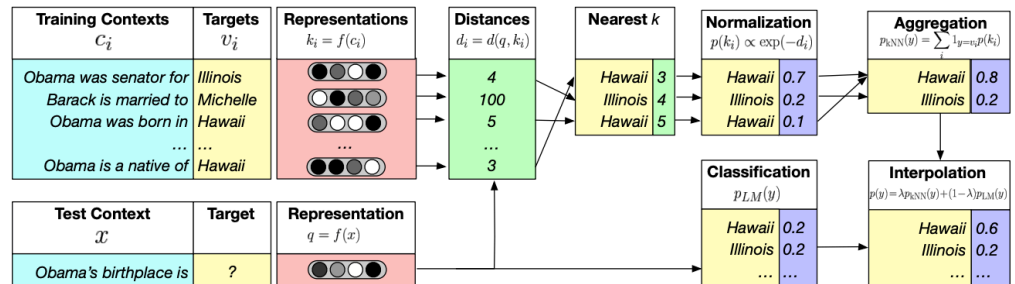
KNN-LM



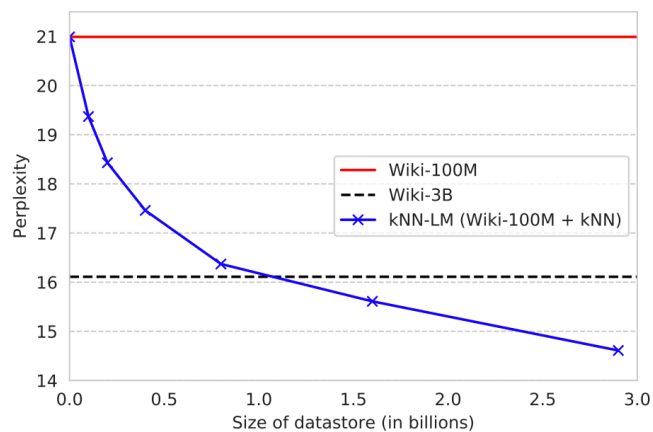
Khandelwal et al (2020)



kNN-LM



kNN-LM



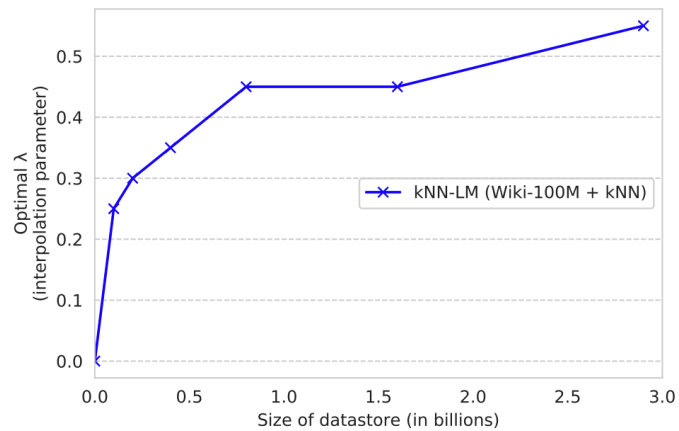


kNN-LM

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Khandelwal et al (2020)

67

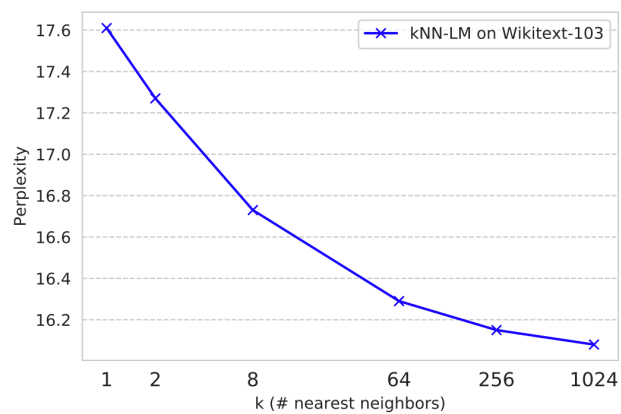


kNN-LM

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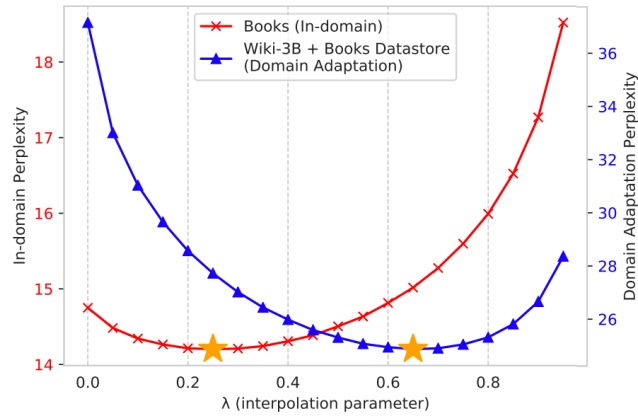


Khandelwal et al (2020)

68



kNN-LM



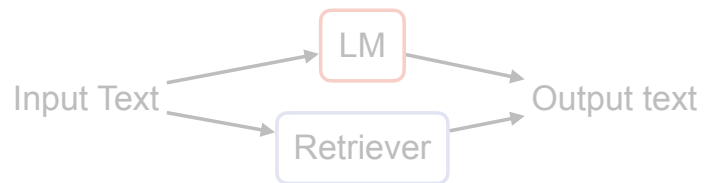
Khandelwal et al (2020)

69

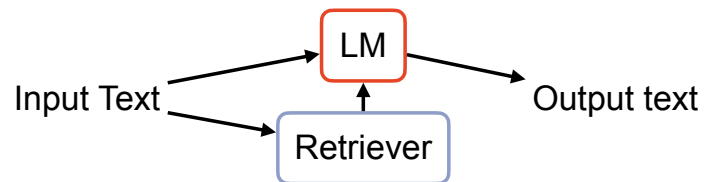


Three general forms

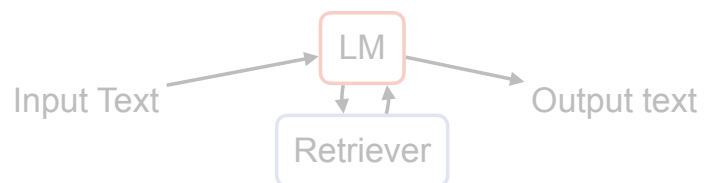
Parallel
Interaction



Sequential
Single
Interaction



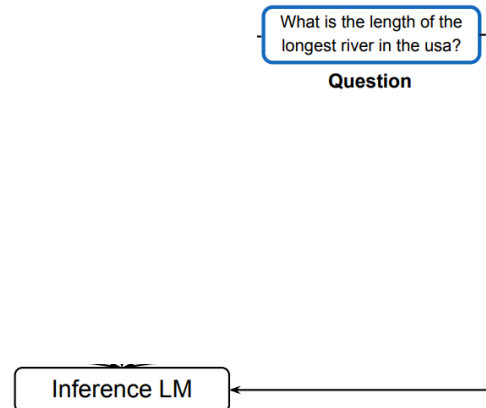
Sequential
Multiple
Interaction



70



Providing examples based on retrieval

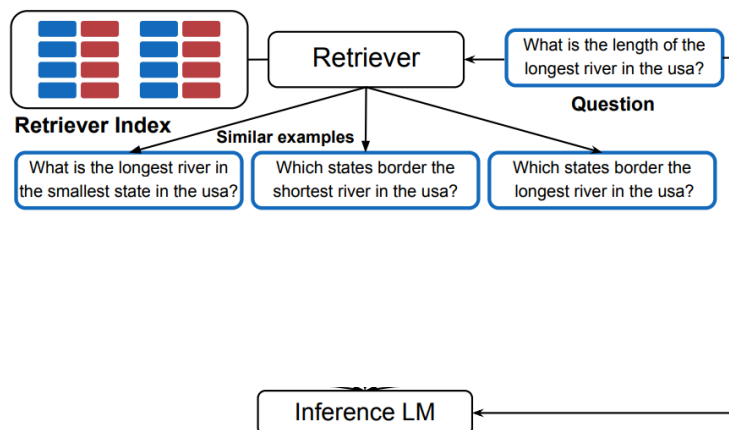


Rubin et al (2022)

71



Providing examples based on retrieval

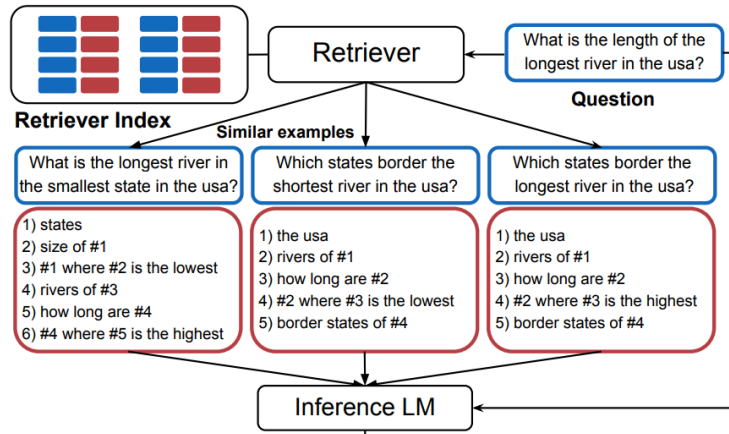


Rubin et al (2022)

72



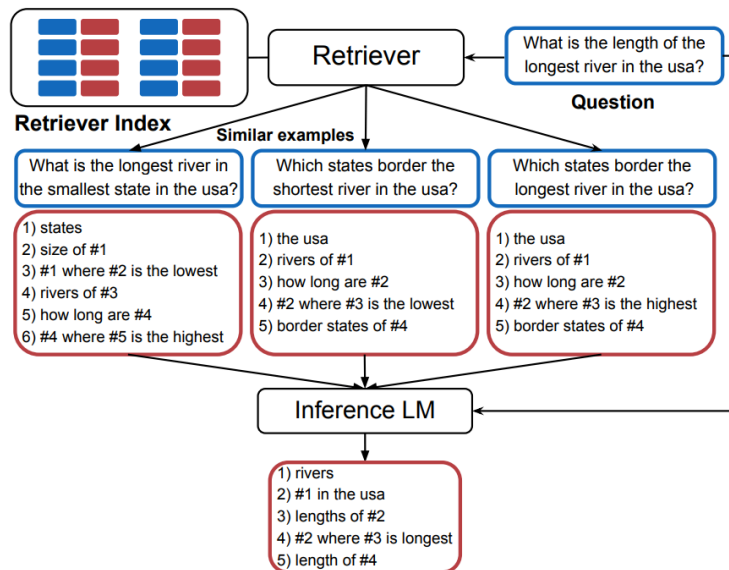
Providing examples based on retrieval



Rubin et al (2022)



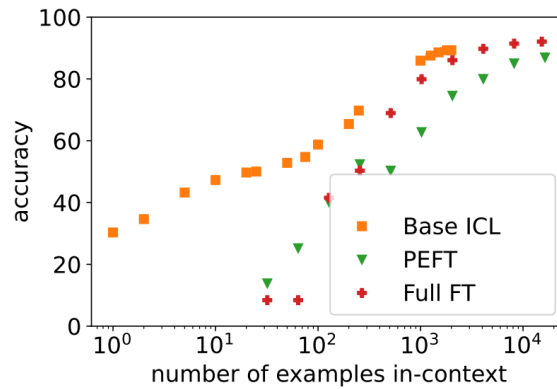
Providing examples based on retrieval



Rubin et al (2022)



How does this interact with the number of examples?



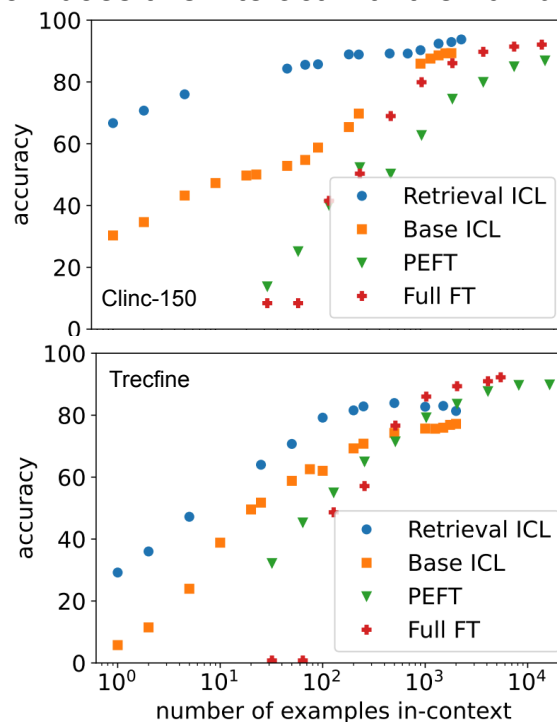
Retrieval ICL:
Find the most relevant
examples in your
training data and use
them in the prompt.

Bertsch et al (2024)

75



How does this interact with the number of examples?



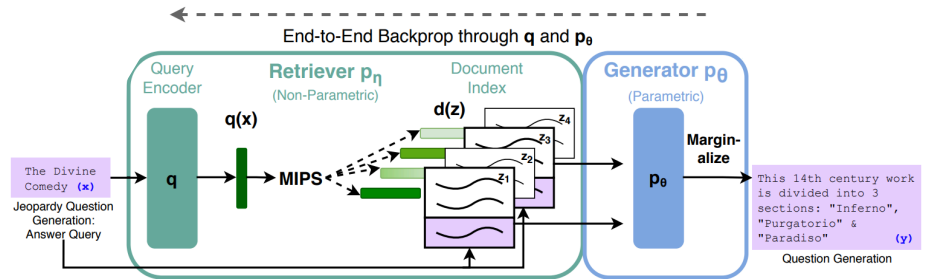
Retrieval ICL:
Find the most relevant
examples in your
training data and use
them in the prompt.

Bertsch et al (2024)

76



RAG: Retrieval Augmented Generation



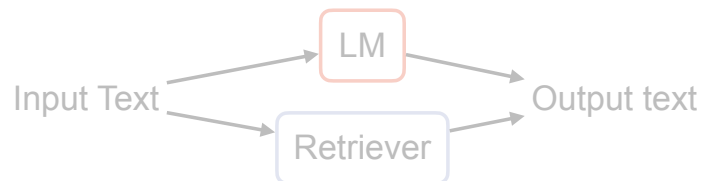
Lewis et al (2020)

77

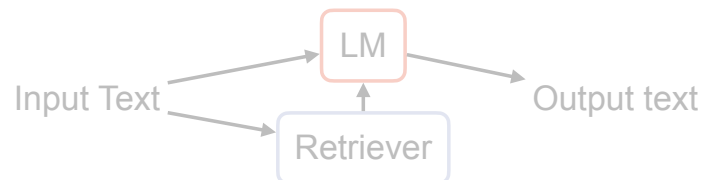


Three general forms

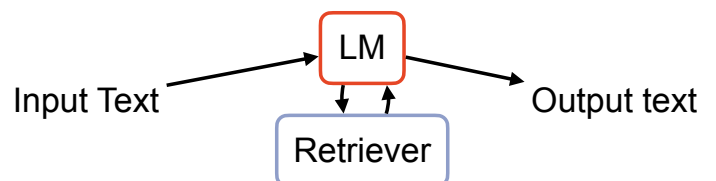
Parallel
Interaction



Sequential
Single
Interaction



Sequential
Multiple
Interaction



78



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FLARE

Input x | Generate a summary about Joe Biden.

Joe Biden (born November 20, 1942) is the 46th president of the United States.

Joe Biden (born November 20, 1942) is the 46th president of the United States.

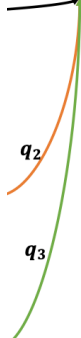
Joe Biden attended the University of Pennsylvania, where he earned a law degree.

He graduated from the University of Delaware in 1965 with a Bachelor of Arts in history and political science.

Joe Biden announced his candidacy for the 2020 presidential election on August 18, 2019.

Joe Biden announced his candidacy for the 2020 presidential election on April 25, 2019.

Retriever



Jiang et al (2023)

79



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FLARE

Joe Biden attended the University of Pennsylvania, where he earned a law degree.

implicit query
by masking

explicit query by
question generation

Joe Biden attended , where he earned .

Ask a question to which the answer is "the University of Pennsylvania"
Ask a question to which the answer is "a law degree"



LM such as ChatGPT

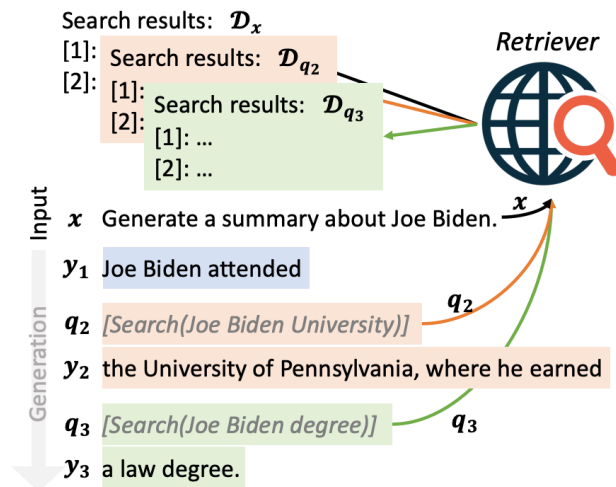
What university did Joe Biden attend?
What degree did Joe Biden earn?

Jiang et al (2023)

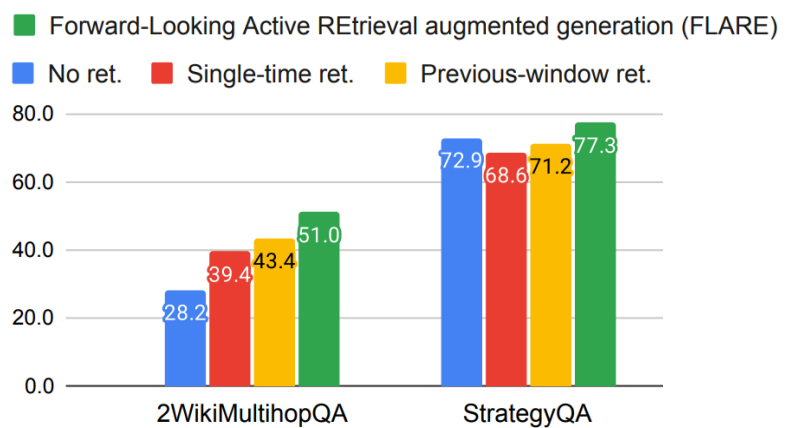
80



FLARE



FLARE





Recap: Retrieval Augmentation

General Idea: Retrieve data from a text collection or database and then use that in the generation process somehow.

Retrieval: Most common approach is to calculate similarity of vector representations between items in the database and some aspect of input.

Use: A variety of methods! Good to know about:

- Treating retrieval as an LM itself and then combining predictions, like an ensemble
- Provide the retrieved content in the prompt
- Interactive methods (related to Agents, which we will see later)

3 minute Break - stretch and visit Menti

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Flies



[I don't know about houseflies, but we definitely caught a lot of fruit flies with our vinegar bowl. Hooray science!]

Source: <https://xkcd.com/357/>



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85



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HuggingFace - a library for various NLP / ML tasks

Pre-work - None this week

86



COMP 4446 / 5046
Lecture 9

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